APRIL-MAY 2026

Large Language Models & Generative AI

From Pytorch to Frameworks

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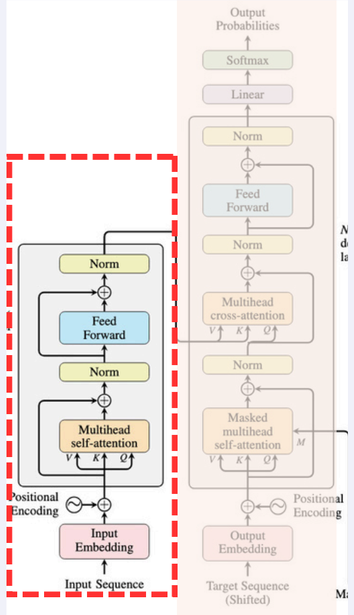
Language Models

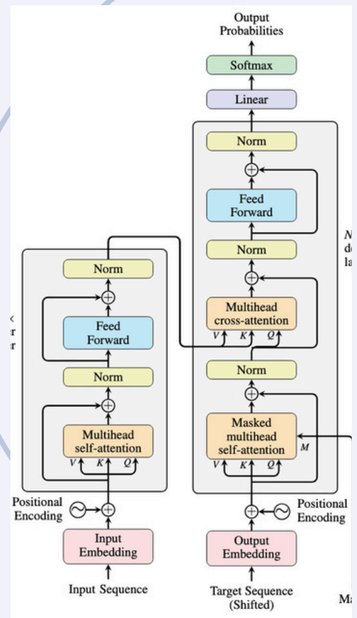
LLM

Encoder-only



Processes the entire input simultaneously using bidirectional self-attention, building a contextual representation of each token.
Ideal for **classification, named entity recognition, and semantic search**.

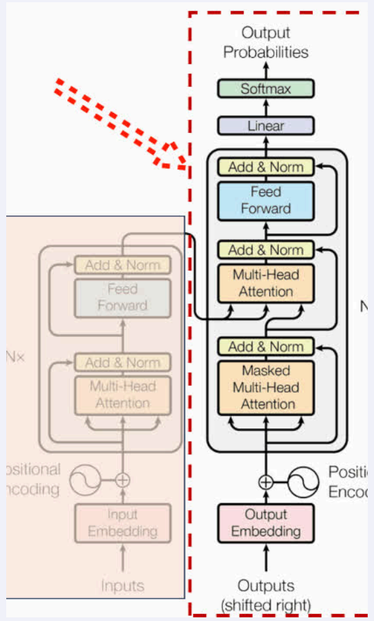
Encoder-decoder



Processes the entire input using the encoder, then generates an output with the decoder based on that representation.

Ideal for **translation or summarization.**

Decoder-only



Uses a single autoregressive stack that generates each token based on all previous tokens, treating both the input and the output as one continuous sequence.

Ideal for **open-ended text generation**

DECODER-ONLY

Uses a single autoregressive stack that generates each token based on all previous tokens: input and output are treated as one continuous sequence.

Input tokens
↓
Embedding + Positional Encoding
↓
Transformer Decoder × N
(masked self-attention + feedforward)
↓
Next token prediction
↓
Repeat until [EOS]

WHEN TO USE:

- **Open-ended text generation**
- **Instruction following (chat)**
- **Code generation**
- **Reasoning tasks**

- ✗ **Classification:** use encoder-only
- ✗ **Translation / summarization:** use encoder-decoder

How decoder-only works

At each step the model sees the entire history and predicts only the next token.
The same causal mask used in the encoder-decoder decoder, but applied to the full sequence

Autoregressive generation: A model is autoregressive when it generates one token at a time, using all previous tokens as context for the next prediction.

Causal Language Modeling (CLM): The pre-training objective of decoder-only models. "Causal" means the model can only see past tokens, never future ones.

Autoregressive generation

Input: "The cat sat on the"

Step 1: model sees "The cat sat on the"
→ predicts "mat"

Step 2: model sees "The cat sat on the mat"
→ predicts "."

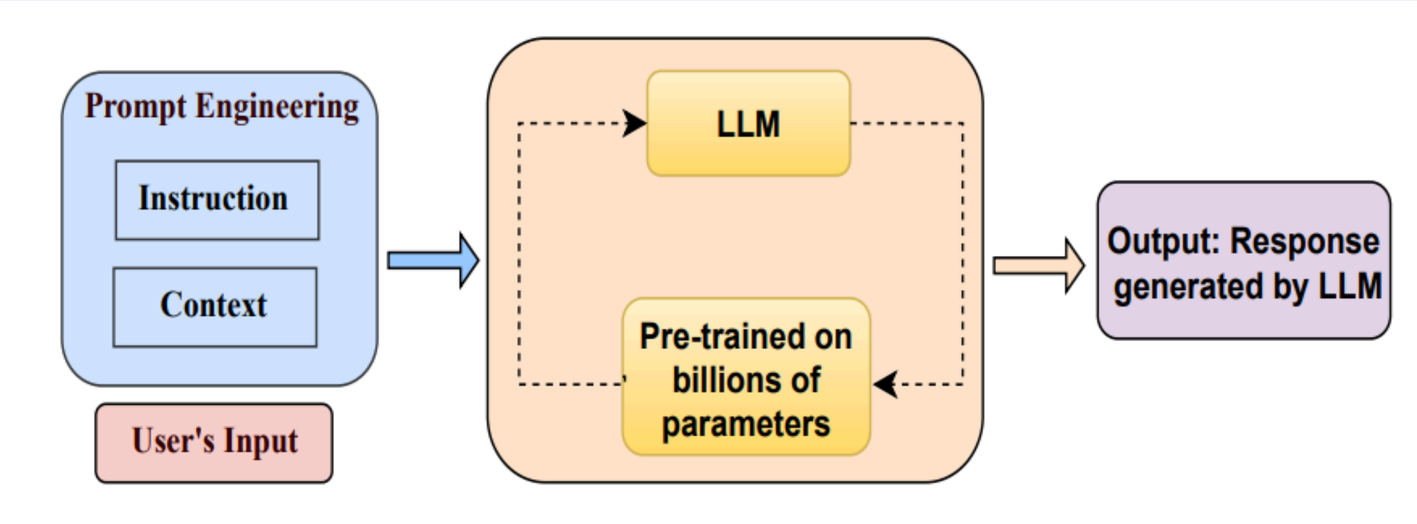
Step 3: model sees "The cat sat on the mat ."
→ predicts [EOS] → stop

Output: "The cat sat on the mat."

Prompt Engineering

Prompt engineering is the process of creating the best **template** for each task the model is expected to perform.

GOAL: enhance the capabilities and guide the LLMs without modifying their core parameters.



PROMPT

Input provided to a language model. Prompts can be natural language instructions or learned vector representations .

Prompting Strategies: a possible taxonomy

- **New Tasks Without Extensive Training** §2.1
  - **Zero-shot Prompting** [Radford *et al.*, 2019]
  - **Few-shot Prompting** [Brown *et al.*, 2020]
- **Reasoning and Logic** §2.2
  - **Chain-of-Thought (CoT) Prompting** [Wei *et al.*, 2022]
  - **Automatic Chain-of-Thought (Auto-CoT)** [Zhang *et al.*, 2022]
  - **Self-Consistency** [Wang *et al.*, 2022]
  - **Logical CoT (LogiCoT) Prompting** [Zhao *et al.*, 2023]
  - **Chain-of-Symbol (CoS) Prompting** [Hu *et al.*, 2023]
  - **Tree-of-Thoughts (ToT) Prompting** [Yao *et al.*, 2023a]
  - **Graph-of-Thought (GoT) Prompting** [Yao *et al.*, 2023b]
  - **System 2 Attention Prompting** [Weston and Sukhbaatar, 2023]
  - **Thread of Thought (ThoT) Prompting** [Zhou *et al.*, 2023]
  - **Chain of Table Prompting** [Wang *et al.*, 2024]

New Task Without Extensive Training

ZERO-SHOT

LLMs are asked to perform a task ***without examples***.

The model receives a **task description** in the prompt and leverages its pre-existing knowledge.

Prompt:
Classify the text into neutral, negative or positive.
Text: I think the vacation is okay.
Sentiment:

FEW-SHOT

LLMs receive a ***few input-output examples*** to induce an understanding of a given task.

Prompt:
Classify the following sentences:
This is awesome! // Negative
This is bad! // Positive
Wow that movie was rad! // Positive
What a horrible show! //

Reasoning and logic: Chain of thought

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

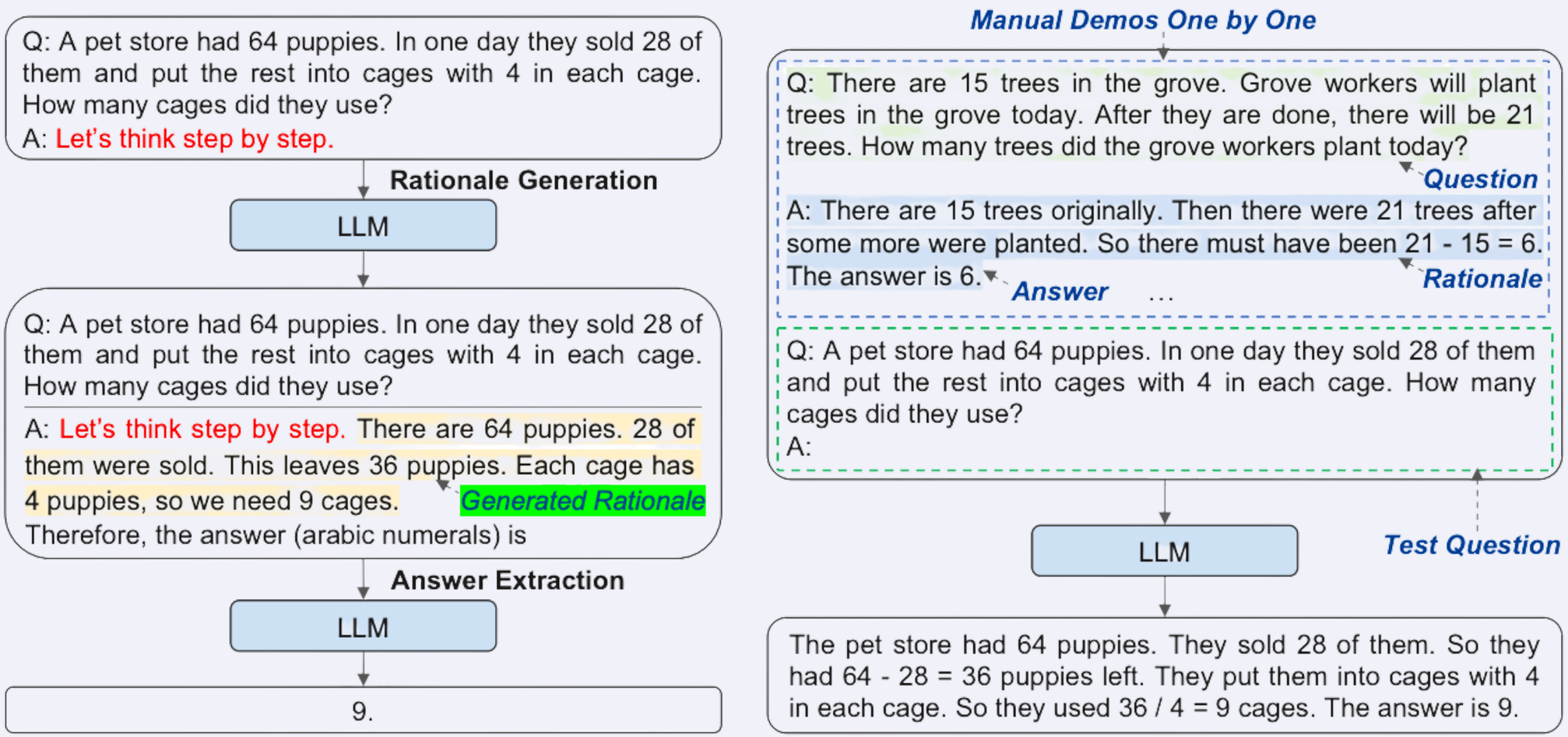
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

Reasoning and logic: Automatic Chain of thought



(a) Zero-Shot-CoT

(b) Manual-CoT

Role-play prompting

Augment the reasoning capabilities of LLM by placing the model into a well-defined role that is related to the domain or task to be solved [10].

The **assigned role** offers a context about the model's identity and expertise.

Result: LLM produces responses that are more natural, consistent, and aligned with the expectations of that role.

Input

User: From now on, you are an excellent math teacher and always teach your students math problems correctly. And I am one of your students.

Assistant: That's great to hear! As your math teacher, I'll do my best to explain mathematical concepts correctly so that you can understand them easily. Feel free to ask any math problems or questions you have, and I'll be glad to assist you. Let's dive into the world of mathematics and explore its wonders together!

User: Xavier was 4 feet tall and grew 3 inches. Cole was 50 inches tall and grew 2 inches over the summer. What is the difference between Cole and Xavier's height now?

Output

Assistant: … Since 1 foot is equal to 12 inches, Xavier's initial height: 4 feet * 12 inches/foot = 48 inches. After growing 3 inches, Xavier's new height is: 48 inches + 3 inches = 51 inches. … Cole's new height after growing 2 inches: 50 inches + 2 inches = 52 inches. … We subtract Xavier's height from Cole's height: 52 inches - 51 inches = 1 inch. Therefore, the difference between Cole and Xavier's height now is 1 inch. ✓

Limitations of LLMs

Hallucinations: generation of incorrect information. That means they can generate statements that are fluent and internally consistent but not grounded in reality.

Lack of true understanding: LLMs do not have direct access to the physical world, sensory experience, or grounded goals. They can model language about the world without having world understanding in the human sense.

Bias and fairness: training data reflects society, and society contains bias, thus models can reproduce or even amplify stereotypes, uneven treatment of groups, and harmful associations.

Context window: LLMs only “see” a finite number of tokens at once. Anything outside the context window is effectively invisible unless summarized or retrieved again.

Computational resources: Modern LLMs are expensive to train and can also be costly to serve at scale (latency, throughput, memory).